Informatica® Cloud Application Integration
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Java Messaging Service Connector Guide

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Preface

Read this connector guide to learn how to set up and use the connection. This guide assumes that you have an understanding of the connector concepts.

CHAPTER 1

JMS connectors overview

JMS Connector replaces the traditional method of creating a messaging service in the Application Integration Console and then establishing the connection between Java applications and messaging brokers.

JMS Connector offers the following benefits:

- **Decoupling**: Enables applications to operate independently, improving scalability and reliability.
- **Reliability**: Provides guaranteed message delivery and supports transactional messaging.
- **Flexibility**: Supports various messaging patterns, including point-to-point, publish-subscribe, and request-response.

Introduction to JMS Connector

A JMS Connector is a middleware component that enables communication between different systems, applications, or services using the JMS API.

JMS Connector facilitates communication between Java applications and messaging systems, such as queues or topics. You can use JMS Connector in enterprise environments to integrate distributed systems and to establish reliable, asynchronous message communication. JMS Connector replaces the traditional method of creating a messaging service in the Application Integration Console and then establishing the connection between Java applications and messaging brokers.

The JMS source service produces events by receiving messages from a JMS provider and publishing those messages as events. When a JMS Service object is initialized, it instantiates its own JMS Connection Factory.

CHAPTER 2

JMS connections overview

Use a JMS connection to connect to a message broker, such as Apache ActiveMQ, IBM MQ, or Oracle WebLogic Server, to send and receive messages.

Basic connection properties

The following table describes the basic properties that you can configure on the **Properties** tab of the connection creation page:

Property	Description
Name	Required. Unique name for the JMS connection that identifies it in the Process Designer. The name must start with an alphabet and can contain only alphabets, numbers, and hyphens (-).
Location	The location of the project or folder where you want to save the connection. Click **Browse** to select a location. If the **Explore** page is currently active and a project or folder is selected, the default location for the connection is the selected project or folder. Otherwise, the default location is the location of the most recently saved asset.
Description	Optional. Description of the connection.
Type	Required. The type of connection you want to use for the connector or service connector. Select **JMS**.
Runtime Environment	Required. The runtime environment for the connection. You can run the connection only on the Secure Agent.
Connection Test	Indicates whether the connection test was successful or not.
OData-Enabled	Not supported for JMS.

JMS connection properties

The following table describes the additional property that you must configure in the **Connection Properties** section:

Property	Description
Hostname	The name of the server or the host where the JMS broker or provider runs along with the protocol that the broker uses. For example, tcp://localhost:61616.
Port	The port for the specific JMS provider or broker.
Client ID	The unique identifier for a JMS client, which is used to identify the client when connecting to a JMS broker or provider. The ID is only required for durable topic subscriptions.
Connection Factory Name	The name of the JMS connection factory that is used to create connections to the associated JMS provider, or JMS topics for publish and subscribe messaging.
Username	The user name to log on to the JMS server.
Password	The password to log on to the JMS server.
Initial context properties	Enter the additional JNDI initial context properties in ampersand-separated, key=value pairs.

JMS source properties

You can configure an event source in an JMS connection.

You can configure an event source in a JMS connection and publish the connection on a Secure Agent. You can then access the event source in a process and deploy the process on a Secure Agent to consume the events generated by the event source.

To create event sources for a JMS connection, click **Add Event Source** on the **Event Sources** tab. Select the event source type as **JMS Reader**. You can add one or more event sources for each JMS connection that you create.

The following table describes the basic event source properties that you can configure:

Property	Description
Name	Required. The event source name that appears in the Process Designer. The name must be unique for the connection. The name must start with an alphabet and can contain only alphabets, numbers, or hyphens (-).
Description	Optional. A description for the JMS event source that appears in the Process Designer.
Enabled	Select **Yes** to make the event source available immediately after it is published. Select **No** to disable the event source until you are ready to use it. Default is **Yes**.

The following table describes the advanced event source properties that you can configure:

Property	Description
Source Type	Required. Select the type of JMS source, such as the queue or topic that the connector will consume messages from.
Queue/Topic Name	Required. Enter the queue or topic name from which the message needs to be consumed.
Concurrent consumers	Select the default number of threads or connections to consume messages from a JMS source, such as a JMS queue or topic. The default number of threads is 1.
Durable Subscription	Enable use durable subscription to allow a subscriber to receive messages even when the subscriber is offline or disconnected. The connection name and the event consumer name are used together to form the unique identifier for the subscription name.
Content Format	Required. Select the JMS message content type to be processed. You can select **Text**, **XML**, or **JSON**. The default content format is text.

The following table describes the other settings that you can configure:

Property	Description
Other Attributes	Enter a URL-encoded list of other broker-specific attributes that you want to configure for the event source. Use the & symbol as the separator.

JMS target properties

After you define a target for a JMS connection, you can publish the connection on a Secure Agent. You can then access the event target in a process and deploy the process on a Secure Agent to publish the events generated by the event target downstream.

To create event targets for a JMS connection, click **Add Event Target** on the **Event Target** tab. Select the event target type as **JMS Writer**. You can add one or more event targets for each JMS connection that you create.

The following table describes the basic event target properties that you can configure:

Property	Description
Name	Required. The event target name that appears in the Process Designer. The name must be unique for the connection. The name must start with an alphabet and can contain only alphabets, numbers, or hyphens (-).
Description	Optional. A description for the JMS event target that appears in the Process Designer.

The following table describes the advanced event target properties that you can configure:

Property	Description
Target Type	Required. The JMS destination, such as the queue or topic that the connector will send messages to.
Queue or Topic Name	Required. The name of the queue or the topic to which the messages need to be sent.
Priority	Select the priority of the message that needs to be sent. 1 is the lowest priority and 9 is the highest priority.
Dynamic Message QoS	Select **Yes** to dynamically override the Quality of Service (QoS) at run time using headers, such as JMSPriority, JMSDeliveryMode, and JMSExpiration.
Time To Live	Enter the maximum time duration in milliseconds for which a message will be retained by the JMS broker before the message is discarded. Default value is -1. -1 indicates that the message will never expire.
Content Format	Required. The message content type that needs to be processed. You can select **Text**, **XML**, or **JSON**. The default content format is text.

The following table describes the other settings that you can configure:

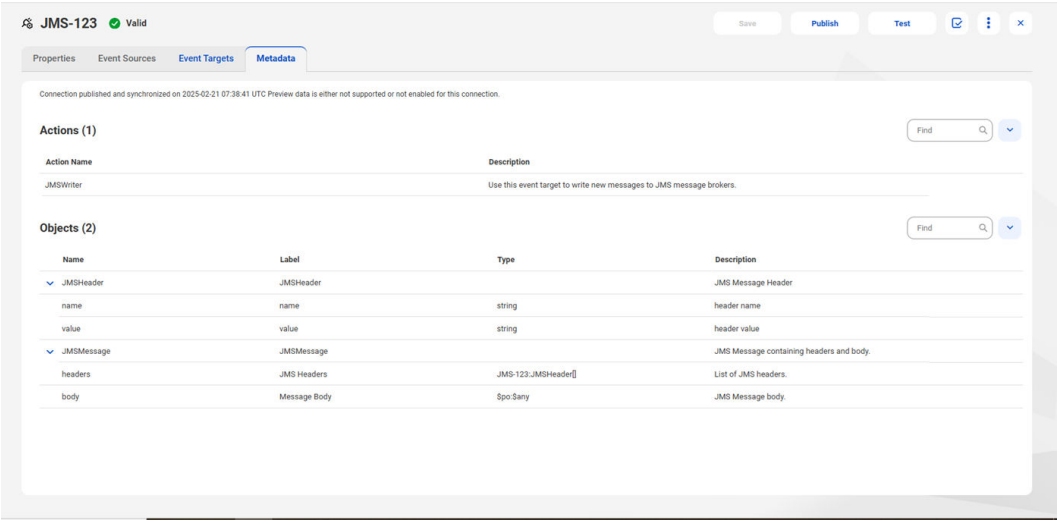
Property	Description
Other Attributes	Enter a URL-encoded list of other broker-specific attributes that you want to configure for the event target. Use the & symbol as the separator.

JMS connection Metadata

After you create a JMS connection, you can validate and save the connection. You can then publish the connection and view the generated process objects on the **Metadata** tab.

When you publish a JMS connection, a list of Actions and Objects appear on the **Metadata** tab.

The following image shows the **Metadata** tab for a published JMS connection:

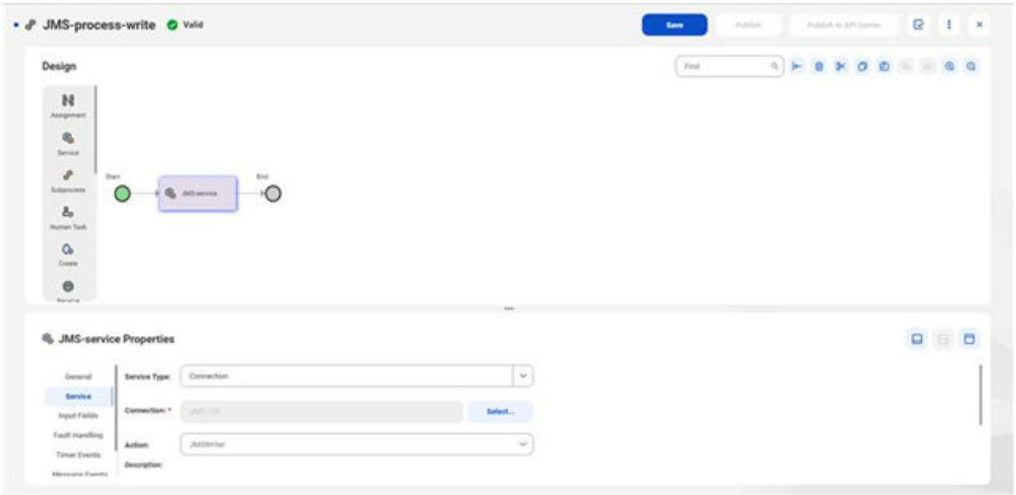


You can call the actions from a process. You can pass the JMS header and message using the JMSHeader and JMSMessage objects.

Using a JMS connection in a process

After you create an JMS connection, you can use it in a Service step in a process.

1. Add a Service step to the process.
2. Click the **Service** tab.
3. From the **Service Type** list, select **Connection**.
4. From the **Connection** list, browse and select the an Azure Event Hubs connection that you created.
5. From the **Action** list, select the action that you want to perform on JMS Connector. For example, select **JMSWriter** to write new messages to JMS message brokers as shown in the following image:



6. Configure further steps as needed.
7. Validate, save, and publish the process.